

```
In [1]: import pandas as pd
import numpy as np
from statsmodels.stats.proportion import proportions_ztest
```

```
In [2]: df = pd.read_csv("online_shoppers_intention.csv")
```

```
In [3]: df.head()
```

```
Out[3]:
```

|   | Administrative | Administrative_Duration | Informational | Informational_Duration | ProductRelated | Prc |
|---|----------------|-------------------------|---------------|------------------------|----------------|-----|
| 0 | 0              | 0.0                     | 0             | 0.0                    | 1              |     |
| 1 | 0              | 0.0                     | 0             | 0.0                    | 2              |     |
| 2 | 0              | 0.0                     | 0             | 0.0                    | 1              |     |
| 3 | 0              | 0.0                     | 0             | 0.0                    | 2              |     |
| 4 | 0              | 0.0                     | 0             | 0.0                    | 10             |     |



```
In [6]: #a/b testing for Weekend
result = df.groupby('Weekend').agg(
    visitors = ('Revenue', 'count'),
    purchases = ("Revenue", 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result['visitors']*100
print(result)
```

|   | Weekend | visitors | purchases | conversion_rate |
|---|---------|----------|-----------|-----------------|
| 0 | False   | 9462     | 1409      | 14.891144       |
| 1 | True    | 2868     | 499       | 17.398884       |

```
In [11]: visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z\_test value is: -3.2529732781407104  
The p-value is : 0.0011420423637110938  
The Alternate Hypothesis is Statistically Correct

```
In [17]: df['VisitorType'].value_counts()
```

```
Out[17]:
```

| VisitorType       |       |
|-------------------|-------|
| Returning_Visitor | 10551 |
| New_Visitor       | 1694  |
| Other             | 85    |

Name: count, dtype: int64

```
In [18]: #Since the 'Other' category in vistortype is ver small we will drop it
df_ab = df[df['VisitorType'] != 'Other']
```

```
In [19]: df_ab['VisitorType'].value_counts()
```

```
Out[19]: VisitorType
Returning_Visitor    10551
New_Visitor          1694
Name: count, dtype: int64
```

```
In [21]: # a/b testing for VisitorType : 'NewVisitor' vs 'Returning Visitor'
result = df_ab.groupby("VisitorType").agg(
    visitors = ("Revenue", 'count'),
    purchases = ("Revenue", "sum")
).reset_index()
result['conversion_rate'] = result['purchases']/result['visitors']*100
print(result)
```

|   | VisitorType       | visitors | purchases | conversion_rate |
|---|-------------------|----------|-----------|-----------------|
| 0 | New_Visitor       | 1694     | 422       | 24.911452       |
| 1 | Returning_Visitor | 10551    | 1470      | 13.932329       |

```
In [22]: visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z\_test value is: 11.605314290218352  
The p-value is : 3.8726217006968784e-31  
The Alternate Hypothesis is Statistically Correct

```
In [23]: #a/b testing for ProductRelated pages
#segmenting the productRelated pages into HIGH and LOW
# We are using median because it will help to segment the data into two groups based on the median
product_related = df['ProductRelated'].median()
df['product_group'] = np.where(
    df['ProductRelated'] >= product_related,
    'High',
    'Low'
)
```

```
In [26]: result = df.groupby('product_group').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

|   | product_group | visitors | purchases | conversion_rate |
|---|---------------|----------|-----------|-----------------|
| 0 | High          | 6240     | 1339      | 21.458333       |
| 1 | Low           | 6090     | 569       | 9.343186        |

```
In [27]: visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z\_test value is: 18.597153510726894  
The p-value is : 3.3882790612577407e-77  
The Alternate Hypothesis is Statistically Correct

```
In [30]: #a/b testing for Product Related Duration
# Creating 'High' and 'Low' categories for Product Related Duration
median_duration = df["ProductRelated_Duration"].median()
df['duration_groups'] = np.where(
    df['ProductRelated_Duration'] >= median_duration,
    'High',
    'Low'
)
result = df.groupby('duration_groups').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

|   | duration_groups | visitors | purchases | conversion_rate |
|---|-----------------|----------|-----------|-----------------|
| 0 | High            | 6165     | 1374      | 22.287105       |
| 1 | Low             | 6165     | 534       | 8.661800        |

```
In [31]: visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z\_test value is: 20.916841464509773  
The p-value is : 3.7622112071127655e-97  
The Alternate Hypothesis is Statistically Correct

```
In [35]: # a/b testing for bounce rates
# creating 'High' and 'Low' categories for Bounce Rates
median_bounce = df['BounceRates'].median()
df['bounce_groups'] = np.where(
    df['BounceRates'] >= median_bounce,
    'High',
    'Low'
)
result = df.groupby('bounce_groups').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

|   | bounce_groups | visitors | purchases | conversion_rate |
|---|---------------|----------|-----------|-----------------|
| 0 | High          | 6165     | 704       | 11.419303       |
| 1 | Low           | 6165     | 1204      | 19.529603       |

```
In [36]: visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z\_test value is: -12.450500871732007  
The p-value is : 1.3897028573513229e-35  
The Alternate Hypothesis is Statistically Correct

